Commodities and Money, working research notes for a mathematical reconstruction of *Capital*, Volume I, Part I

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Abstract

We begin a chapter-by-chapter mathematical research program on Marx's *Capital*. This installment concerns Volume I, Part I, Chapters 1–3. It separates physical quantities, labor-value magnitudes, value expressions, and monetary realization. A finite production network supplies one explicit closure for socially necessary labor coefficients; a graph formalizes consistent quantitative expressions; a commodity-money standard converts labor magnitudes into monetary magnitudes; and a constrained stochastic exchange process separates a quoted price from an accomplished sale. We prove conditional results on production, exchange consistency, conservation, and realization delay, and derive stochastic comparative statics. These are results of the stated models, not proofs of Marx's theory or of historically inevitable crises.

Contents

1 Scope, sources, and method **2**

2 Chapter 1: quantities and the two aspects of labor **3**
2.1 Goods are typed quantities, not interchangeable numbers 3
2.2 Concrete activity and a social reduction operator 3
2.3 A simultaneous production closure 4

3 Chapter 1: value expressions and their quantitative geometry **5**
3.1 The simple expression is oriented 5
3.2 When do ratios have a common scalar representation? 5
3.3 A common equivalent needs an institution 5

4 Chapter 1: a limited representation of fetishism **6**

5 Chapter 3: labor magnitudes and monetary expression **6**
5.1 Commodity-money standard and value prices 6
5.2 Quoted prices and realized receipts 6

6 Chapter 2 and Chapter 3: a stochastic exchange model **7**
6.1 State and admissible events 7
6.2 The two acts of simple circulation 8
6.3 Circulation, hoarding, and a finite accounting identity 9

7 Stochastic production coefficients and value prices **9**
7.1 A precise local domain 9
7.2 Itô formula for network labor coefficients 9
7.3 Relative prices when money also has a changing labor value 10

8 Worked examples and diagnostic counterexamples **10**
8.1 Two reproducible goods and an external money commodity 10
8.2 Cycle inconsistency does not identify its social cause 11
8.3 Money available in aggregate need not be available to a buyer 11

9 Bridge to the existing mathematical-finance research **11**
9.1 What this installment provides 11
9.2 Finite-horizon measure changes: an explicit check 11

10 Open closures and the next installments **12**

1 Scope, sources, and method

The primary textual sequence is Chapter 1 (commodities), Chapter 2 (exchange), and Chapter 3 (money) [1, 2, 3]. Chapter and section references are used because pagination varies between editions. The library copy `Capital-Volume-I.pdf` identifies the Moore–Aveling translation edited by Engels and supplies a local reference edition.

The purpose is a reconstruction with explicit commitments. A *textual category* is something to interpret; an *assumption* selects an operational representation; a *proposition* follows from that representation. We do not infer that a category has been explained merely because a variable has been assigned to it. In particular, neither the historical institution of money nor the reduction of heterogeneous labor is derived by an algebraic change of units.

The existing *Stochastic Marxian Macroeconomics* draft uses $\lambda = \ell(I - A)^{-1}$ and a monetary labor-time conversion in its production network. Here we examine the assumptions behind that earlier notation before introducing surplus-value, wages, accumulation, or financial claims. Those categories require later installments.

Textual location	Category to retain	Mathematical representation here
Ch. 1, §§1–2	Useful goods; concrete and abstract labor	Typed quantities; labor-reduction coefficients; a conditional production closure
Ch. 1, §3	Relative and equivalent forms	Oriented value expressions; consistent edge ratios; a designated common equivalent
Ch. 1, §4	Fetishism	A many-to-one map from social data to observable magnitudes; an explicit limit of the representation
Ch. 2	Owners and exchange	Ownership inventories and admissible bilateral transfers
Ch. 3, §§1–2	Measure and circulation	Commodity-money standard; quoted prices; completed sales and monetary balances
Ch. 3, §3	Hoards, payment, world money	A circulation extension and an agenda for contracts and international settlement

2 Chapter 1: quantities and the two aspects of labor

2.1 Goods are typed quantities, not interchangeable numbers

Fix n reproducible commodity types and H owners. A type includes a quality specification, location and delivery date when relevant, and a physical unit. A physical bundle belongs to

$$X = \prod_{i=1}^{n} \mathbb{R}_+ \, e_i,$$

where e_i labels the unit of type i. Owner a holds $x^a \in X$. The addition of coordinates records inventories; it does not assert that a meter of cloth is physically comparable to a kilogram of grain.

Let $U_{ai}(q, z) \in \{0, 1\}$ indicate whether q units of type i can serve a relevant use for owner a in social state z. This predicate is not a cardinal utility function. It permits usefulness to depend on quality, infrastructure, and social circumstances. Useful objects need not have labor value, and possession of an object does not guarantee a buyer. The model below covers reproducible products of labor offered for exchange. It does not force land or every priced object into that domain.

2.2 Concrete activity and a social reduction operator

Let $k = 1, \ldots, K$ label labor activities. Suppose L_{kj} is the time of activity k required per unit of output j under the selected normal production conditions. Specify positive reduction weights $\alpha_k(z)$, measured in simple-labor hours per hour of activity k. The resulting direct abstract-labor coefficient is

$$\ell_j(z) = \sum_{k=1}^{K} \alpha_k(z) L_{kj}(z). \tag{1}$$

Concrete activities remain distinct in L; ℓ records their reduction to one accounting magnitude. Linearity and the weights are assumptions. They do not establish the social process by which complex labor is reduced to simple labor, nor explain why privately performed labor acquires social validity.

For a firm f, let h_{fj} be its private hours per unit. We do not set the social coefficient equal to h_{fj}. A producer's inefficiency cannot increase the social magnitude simply by consuming more time. One possible empirical rule is an output-weighted mean over a specified set of normal producers. Other rules select regulating techniques or normal capacity. Choosing between them is a substantive research question; a mean is not supplied by the phrase "socially necessary."

2.3 A simultaneous production closure

Assumption 2.1 (Normal reproducible technology)**.** *Fix physical units and one accounting period. Let $A = (a_{ij}) \in \mathbb{R}_+^{n\times n}$, where a_{ij} is units of input i consumed per unit of output j, and let $\ell \in \mathbb{R}_{++}^{1\times n}$ be direct simple-labor hours per output unit. Assume the spectral radius $\rho(A) < 1$. We use circulating inputs, fixed coefficients, no joint production, and no endogenous technique choice. The coefficients describe socially normal production, not an arbitrary individual workshop.*

Definition 2.2 (Reproduction labor coefficients)**.** *A row vector v of labor hours per commodity unit satisfies*

$$v = vA + \ell. \tag{2}$$

The labor magnitude assigned to bundle q is $V(q) = vq$.

Equation (2) is an input-output reconstruction, not an equation written by Marx in Part I. Its recursive treatment of consumed inputs anticipates analysis of production. It identifies v with socially necessary labor requirements only within the chosen closure. An alternative temporal closure values inputs at their production or acquisition dates; these alternatives need not agree during technical change. We retain the simultaneous closure as a transparent baseline.

Proposition 2.3 (Existence, uniqueness, and positivity)**.** *Under Assumption 2.1, the unique solution is*

$$v = \ell R, \qquad R = (I - A)^{-1} = \sum_{r=0}^{\infty} A^r, \tag{3}$$

and $v_i > 0$ for every i.

Proof. The spectral-radius assumption makes the Neumann series converge to the inverse of $I - A$. Every summand is nonnegative and $R \geq I$ entrywise. Thus $\ell R \geq \ell > 0$. Multiplying (2) on the right by R proves existence and uniqueness. The term ℓA^r counts labor through input chains of length r. □

Proposition 2.4 (Net-product identity)**.** *For a nonnegative gross output x with net product $y = (I - A)x \geq 0$,*

$$vy = \ell x. \tag{4}$$

Proof. Multiply $v(I - A) = \ell$ by x. □

The identity concerns the *net* product. Gross output has labor magnitude $vx = vAx + \ell x$; confusing it with newly performed labor double-counts intermediate inputs. The identity is conditional on the production closure and does not prove that market revenue equals it.

Proposition 2.5 (Comparative statics)**.** *For infinitesimal changes of admissible coefficients,*

$$\delta v = (\delta\ell + v\,\delta A)R. \tag{5}$$

If $A' \leq A$ and $\ell' \leq \ell$ entrywise, with both systems satisfying the assumptions, then $v' \leq v$ entrywise.

Proof. Differentiate $v(I - A) = \ell$. For the finite comparison, induction gives $(A')^r \leq A^r$ for every r. Apply the nonnegative Neumann series. □

The proposition specifies one sense in which greater productivity reduces labor requirements. Technique substitution, quality changes, joint products, and changing labor-reduction weights require new arguments. More physical output need not imply more labor magnitude.

3 Chapter 1: value expressions and their quantitative geometry

3.1 The simple expression is oriented

The expression

$$q_i \text{ units of commodity } i \quad \longmapsto \quad q_j \text{ units of commodity } j$$

places i in the relative role and j in the equivalent role. Under an ideal expression at the labor coefficients above,

$$q_i v_i = q_j v_j, \qquad r_{ij} = q_j/q_i = v_i/v_j. \tag{6}$$

The scalar equality is symmetric; the roles in the expression are not. Store the role assignment as an oriented edge. Algebraic reciprocity $r_{ji} = r_{ij}^{-1}$ does not itself imply reciprocal direct saleability. Real market quotations are introduced separately below.

An expanded expression for one commodity is a family of outgoing edges to several equivalents. A general expression arranges many commodities against a common equivalent g. These graph structures describe quantitative forms; they do not establish a historical sequence of barter stages.

3.2 When do ratios have a common scalar representation?

Consider a connected undirected graph whose edges have strictly positive ratios in each orientation, with $r_{ji} = r_{ij}^{-1}$. The graph here records comparability, not permission to execute a trade.

Proposition 3.1 (Cycle consistency)**.** *There is a positive vector w, unique up to a common positive factor, such that $r_{ij} = w_i/w_j$ on every edge if and only if the product of ratios around every cycle is* 1*.*

Proof. If such w exists, multiplication telescopes around any cycle. Conversely fix a root g and $w_g = 1$. Define w_i as the product of edge ratios along a path from i to g. Any two paths give the same product because their concatenation is a closed walk, whose factors cancel by the cycle condition. For the edge $i \to j$, this gives $w_i = r_{ij} w_j$. Connectivity fixes all coordinates relative to g. □

This theorem does *not* identify w as labor values. Any consistent positive price vector gives such ratios. The labor interpretation enters through (2), independent of graph consistency. For disconnected graphs, there is one undetermined scale per component.

Taking logarithms gives

$$b_{ij} = \log r_{ij} = \phi_i - \phi_j, \qquad \phi_i = \log w_i.$$

Thus edge data must be an exact potential difference. With noisy data, one can estimate ϕ by weighted least squares over the edges, fixing $\phi_g = 0$. Nonzero cycle sums are quantitative inconsistencies to investigate. They need not be executable arbitrage: accessibility, timing, inventory, bid–ask spreads, and transaction costs matter.

3.3 A common equivalent needs an institution

Assumption 3.2 (Money institution)**.** *One commodity g is socially accepted in settlement in a specified domain. It is divisible at the scale of this model, has positive labor coefficient v_g, and a fixed monetary denomination $\nu > 0$, measured in currency units per physical unit of g.*

The choice of g is an institutional assumption. A graph always permits a mathematical root; this is insufficient to explain the selection, acceptance, or durability of money. Commodity money is the baseline appropriate to the text's gold abstraction. Fiat money and bank deposits require a distinct institutional model.

4 Chapter 1: a limited representation of fetishism

Let z contain techniques, labor activities, ownership, contractual rules, and patterns of social acceptance. Observable magnitudes may be represented by a map

$$\Psi : z \longmapsto \big(v(z), p(z)\big).$$

This map is usually many-to-one. For example, with one sector and labor already reduced to simple hours,

$$(A, \ell) = (0, 1) \quad \text{and} \quad (A, \ell) = (1/2, 1/2)$$

both give $v = 1$. Ownership changes can also leave A, ℓ, and a chosen price rule unchanged. Magnitudes alone therefore do not identify the relations represented by the full state.

This non-identification is a useful mathematical warning against treating a price as an intrinsic physical attribute. It is not a full theory of fetishism: a statistical projection cannot by itself explain the objective social power of commodity relations or the appearance those relations assume for participants. An adequate extension must retain ownership and institutions in the state rather than discard them after computing v.

5 Chapter 3: labor magnitudes and monetary expression

5.1 Commodity-money standard and value prices

Under Assumption 3.2, define

$$m = \frac{\nu}{v_g}, \qquad p_i^* = m v_i = \nu \frac{v_i}{v_g}. \tag{7}$$

Here m is currency units per simple-labor hour and p_i^* is currency units per commodity unit. This is a *commodity-standard monetary expression of labor time*; it need not coincide with an aggregate MELT defined from monetary net output and living labor in a different model. If g is included among the n commodities, (7) gives $p_g^* = \nu$. If g is external, its production coefficient must be supplied by a separate system; it cannot be inferred from A.

Fixing a currency unit measures prices; specifying v_g relates that standard to labor magnitudes. A denomination change $\nu \mapsto c\nu$ multiplies all p_i^* and m by c without changing physical quantities or labor coefficients. If all v_i, including v_g, fall by the same factor, the value prices p_i^* remain unchanged even though labor requirements have fallen.

5.2 Quoted prices and realized receipts

Introduce a positive quoted-price vector p independently. For the reproducible labor products in our domain, parameterize

$$p_i = p_i^* \exp(\xi_i). \tag{8}$$

This is a decomposition, not a theory of ξ_i. If g is included, set $\xi_g = 0$ to preserve the denomination. Scarcity, bargaining, institutional price setting, and market access can later determine the wedge.

We do not impose $\sum_i p_i q_i = m \sum_i v_i q_i$ for arbitrary traded bundles; such an aggregate equality needs an additional closure.

If units actually sold during an interval are recorded by finite-variation processes Q_i^{sold}, then gross receipts are

$$R_T^{\mathrm{sales}} = \sum_i \int_{(0,T]} p_{i,t-} \, \mathrm{d}Q_{i,t}^{\mathrm{sold}}. \tag{9}$$

A quote or an inventory marked at that quote is not this cash flow. Products can remain unsold when every displayed p_i is positive.

6 Chapter 2 and Chapter 3: a stochastic exchange model

6.1 State and admissible events

Work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. Owner a holds non-money inventory $x_t^a \in \mathbb{R}_+^{n_c}$ and money balance $b_t^a \geq 0$, where n_c counts non-money types. Production is paused during the exchange interval. Balances are claims to the accepted settlement asset; for the commodity-money baseline, b^a/ν is the physical stock of g owned by a.

A sale event $e = (a, c, i, q)$ means seller a transfers $q > 0$ units of commodity i to buyer $c \neq a$ at the pre-event quote p_i. Its updates are

$$\begin{aligned} x^{a\prime} &= x^a - q e_i, \quad x^{c\prime} = x^c + q e_i, \\ b^{a\prime} &= b^a + p_i q, \quad b^{c\prime} = b^c - p_i q. \end{aligned} \tag{10}$$

It is admissible only if $x_i^a \geq q$, $b^c \geq p_i q$, the buyer can use or pass on the good, and both parties accept the transfer. Those conditions represent ownership and contract restrictions. They do not reduce exchange to an automatic physical motion of objects.

For a finite list of possible lot sizes, let N_t^e count sales of type e with predictable intensity $\kappa_e(t, Z_{t-})$, where Z includes inventories, balances, quotes, and acceptance variables. Set the intensity to zero outside the admissible set. Suppose the total intensity is bounded on each finite time horizon and events do not occur simultaneously. A construction by thinning marked Poisson events then produces a nonexplosive process. Conditional on an exogenous quote path, the exchange jump operator is

$$(\mathcal{J}_t f)(z) = \sum_e \kappa_e(t, z) \big[f(T_e(t, z)) - f(z) \big]. \tag{11}$$

If quotes themselves have Markov dynamics, their generator must be added for a generator of the full state. Formula (11) alone describes the exchange part.

Proposition 6.1 (Nonnegativity and conservation)**.** *The sale process preserves nonnegative holdings and, pathwise,*

$$\sum_a x_{i,t}^a = \sum_a x_{i,0}^a, \qquad \sum_a b_t^a = \sum_a b_0^a. \tag{12}$$

For labor coefficients frozen during the interval, total inventory labor magnitude is also conserved.

Proof. Admissibility prevents a seller or buyer from giving more than it holds. Each update subtracts exactly what it adds to another owner's account. Sum the updates over the finite number of events on each finite horizon. Multiply the physical conservation identity by the fixed coefficients to obtain conservation of inventory labor magnitude. □

Exchange can redistribute money and goods without creating total labor magnitude in this model. It need not preserve an individual owner's labor-equivalent holdings when $p_i \neq p_i^*$. Nor does the proposition exclude changing monetary wealth when quotes move: revaluation is different from production. With changing v_t, inventory conservation does not imply constant $v_t x_t$.

6.2 The two acts of simple circulation

For an owner selling type i to buy type j, let τ_s be the first successful sale and $\tau_b \geq \tau_s$ the subsequent intended purchase. The circuit is

$$C_i \xrightarrow{\tau_s} M \xrightarrow{\tau_b} C_j.$$

The purchase may never occur. Between the acts, money can remain idle. Its purchasing power over j changes with $p_{j,t}$. The two stopping times therefore retain a separation lost in instantaneous barter. This is simple commodity circulation; it does not yet describe the M–C–M' circuit of capital.

Proposition 6.2 (Sale delay under a specified hazard)**.** *Suppose a single lot has no competing sale event and has deterministic, locally integrable first-sale hazard $h(t) \geq 0$. Then*

$$\mathbb{P}(\tau_s > T) = \exp\left(-\int_0^T h(u)\,\mathrm{d}u\right), \qquad \mathbb{E}[\tau_s] = \int_0^\infty \exp\left(-\int_0^t h(u)\,\mathrm{d}u\right)\,\mathrm{d}t, \tag{13}$$

where the expectation may be infinite. For constant $h > 0$, $\mathbb{E}[\tau_s] = 1/h$.

Proof. The survival function solves $S'(t) = -h(t)S(t)$ almost everywhere with $S(0) = 1$. The expectation formula follows by integrating the tail of a nonnegative random variable. □

For $h \equiv 0$, a positive quote gives no realization. Even when $\int_0^\infty h(u)\,\mathrm{d}u = \infty$ guarantees eventual sale almost surely, mean delay can be infinite: $h(t) = 1/(1+t)$ gives $S(t) = 1/(1+t)$. Thus almost-sure realization and timely realization are different.

For a random hazard independent of the exponential trigger, conditional on the hazard path the same survival formula holds, and the unconditional survival is its expectation. That Cox-process formula must not be applied automatically to an endogenous hazard that feeds back on the sale itself. A general predictable intensity gives the compensator identity

$$N_t^e - \int_0^t \kappa_e(u, Z_{u-})\,\mathrm{d}u \quad \text{is a local martingale.} \tag{14}$$

With suitable integrability it is a martingale. Expected receipts can then be calculated from compensators, for example

$$\mathbb{E}[R_T^{\text{sales}}] = \mathbb{E}\left[\sum_e \int_0^T p_{i(e),t-} q_e \kappa_e(t, Z_{t-})\,\mathrm{d}t\right],$$

when the nonnegative sums and integrals are well defined (or integrable for a finite-valued expectation).

6.3 Circulation, hoarding, and a finite accounting identity

Divide accepted money into active balances B_t^{act} and hoards H_t^{hoard}, so $B = B_t^{\mathrm{act}} + H_t^{\mathrm{hoard}}$ in this closed exchange system. Hoarding changes availability and potentially the intensities; it does not destroy the accepted money stock.

Let $F_{[0,T]}$ be the sum of monetary amounts in completed cash sales. For $T > 0$ and mean active balance $\overline{B}^{\mathrm{act}} = T^{-1} \int_0^T B_t^{\mathrm{act}} \, \mathrm{d}t > 0$, define the turnover count $V_T = F_{[0,T]} / \overline{B}^{\mathrm{act}}$. Then

$$F_{[0,T]} = \overline{B}^{\mathrm{act}} V_T. \tag{15}$$

This is an accounting definition. It supplies no causal rule that a money-stock change must cause a proportional price change. Moreover, F is realized sales, not the sum of prices of all goods offered. Credit sales, debt settlement, netting, and cross-border payments require an expanded ledger before they can be included.

7 Stochastic production coefficients and value prices

7.1 A precise local domain

Let W be a d-dimensional Brownian motion under $\mathbb{P}$. On an open domain with $I - A$ invertible, suppose

$$\begin{aligned} \mathrm{d}A_t &= B_t \, \mathrm{d}t + \sum_{k=1}^{d} C_t^k \, \mathrm{d}W_t^k, \\ \mathrm{d}\ell_t &= a_t \, \mathrm{d}t + \sum_{k=1}^{d} c_t^k \, \mathrm{d}W_t^k. \end{aligned} \tag{16}$$

Here B, C^k are matrices and a, c^k are row vectors. Assume adapted coefficients with the usual local integrability for these Itô processes. For the economic interpretation, require $A_t \geq 0$, $\ell_t > 0$, and $\rho(A_t) < 1$.

An unrestricted Brownian coefficient model need not preserve those inequalities. The following calculation is local, up to the first exit from the admissible domain, with localization bounding R_t and the coefficients. A global economic model needs an invariant-domain construction, reflection, or suitable transformations. A bound on $\rho(A)$ alone is not an entrywise bound on A or a uniform bound on the inverse for all nonnormal matrices.

7.2 Itô formula for network labor coefficients

Set $R_t = (I - A_t)^{-1}$ and $v_t = \ell_t R_t$. Differentiating the inverse and using matrix Itô products gives

$$\mathrm{d}R_t = \left(R_t B_t R_t + \sum_k R_t C_t^k R_t C_t^k R_t \right) \mathrm{d}t + \sum_k R_t C_t^k R_t \, \mathrm{d}W_t^k. \tag{17}$$

Consequently

$$\begin{aligned} \mathrm{d}v_t = &\left[a_t R_t + v_t B_t R_t + \sum_k (v_t C_t^k + c_t^k) R_t C_t^k R_t \right] \mathrm{d}t \\ &+ \sum_k (c_t^k + v_t C_t^k) R_t \, \mathrm{d}W_t^k. \end{aligned} \tag{18}$$

To verify the matrix order, start from $(I - A_t)R_t = I$. The diffusion coefficient of R is RC^kR. Its quadratic cross-product with A then yields the drift correction RC^kRC^kR. In $\mathrm{d}(\ell R) = \mathrm{d}\ell\, R + \ell\ \mathrm{d}R + \mathrm{d}\ell\ \mathrm{d}R$, the last term is $\sum_k c^k RC^kR\,\mathrm{d}t$, which supplies the labor–technology covariance correction in (18).

These are exact local identities for the simultaneous closure. They do not supply the coefficients in (16), nor derive a law of technical change from exchange. A stochastic law of motion must follow an economic specification, rather than stand in for one.

7.3 Relative prices when money also has a changing labor value

Suppose positive coefficients v_i, v_g have local dynamics

$$\frac{\mathrm{d}v_i}{v_i} = \beta_i\,\mathrm{d}t + \sigma_i \cdot \mathrm{d}W_t, \qquad \frac{\mathrm{d}v_g}{v_g} = \beta_g\,\mathrm{d}t + \sigma_g \cdot \mathrm{d}W_t, \tag{19}$$

with all coefficients evaluated at t. For fixed ν, the ideal commodity-money price $p_i^* = \nu v_i / v_g$ satisfies

$$\frac{\mathrm{d}p_i^*}{p_i^*} = \left[\beta_i - \beta_g + \|\sigma_g\|^2 - \sigma_i \cdot \sigma_g\right]\mathrm{d}t + (\sigma_i - \sigma_g) \cdot \mathrm{d}W_t. \tag{20}$$

Equivalently,

$$\mathrm{d}\log p_i^* = \left[\beta_i - \beta_g - \tfrac{1}{2}\left(\|\sigma_i\|^2 - \|\sigma_g\|^2\right)\right]\mathrm{d}t + (\sigma_i - \sigma_g) \cdot \mathrm{d}W_t. \tag{21}$$

The inverse-money correction and the covariance term cannot be dropped. Identical proportional dynamics for v_i and v_g leave their ratio constant. If g belongs to the network, its dynamics are the corresponding coordinate of (18); they cannot be chosen independently.

For continuous Itô ξ_i in (8), $\mathrm{d}\log p_i = \mathrm{d}\log p_i^* + \mathrm{d}\xi_i$. Converting to $\mathrm{d}p_i/p_i$ requires the quadratic variation of this entire sum, including cross covariation. This is one route to a model of price fluctuations anchored in production, but neither p_i^* nor p_i is assumed to be a risk-neutral asset price.

8 Worked examples and diagnostic counterexamples

8.1 Two reproducible goods and an external money commodity

Take

$$A = \begin{pmatrix} 1/10 & 1/5 \\ 1/5 & 1/10 \end{pmatrix}, \qquad \ell = (2/5, 4/5).$$

The eigenvalues are $3/10$ and $-1/10$, so the assumptions hold. Direct calculation gives

$$R = \frac{1}{77}\begin{pmatrix} 90 & 20 \\ 20 & 90 \end{pmatrix}, \qquad v = (52/77, 80/77).$$

For net output $y = (10, 5)^\top$,

$$x = Ry = (1000/77, 650/77)^\top, \qquad vy = \ell x = 920/77 \text{ labor hours.}$$

Specify an external money commodity with $v_g = 1/2$ hour per unit and $\nu = 10$ currency units per unit. Then $m = 20$ and

$$(p_1^*, p_2^*) = (1040/77, 1600/77) \approx (13.5065, 20.7792).$$

The ideal ratio is $r_{12} = 13/20$. Halving ℓ with fixed A and fixed v_g halves v and these prices. Halving both ℓ and v_g instead leaves these prices unchanged. The money commodity's production specification is additional data in this example.

8.2 Cycle inconsistency does not identify its social cause

Three observed oriented ratios $r_{12} = 2$, $r_{23} = 3$, and $r_{31} = 1/5$ have cycle product $6/5$. No scalar vector reproduces all three simultaneously. This diagnoses failure of the scalar representation; it does not tell us whether the observations differ in delivery date, quality, access, or bargaining conditions.

8.3 Money available in aggregate need not be available to a buyer

Owners A, B, C begin with money balances $(6, 0, 0)$. Suppose B owns one unit of good 1 and C one unit of good 2, each priced at 6. First A buys good 1 from B, producing balances $(0, 6, 0)$. Then B buys good 2 from C, producing $(0, 0, 6)$. Sales total 12 using a stock of 6. Reversing the order is infeasible because B initially has no money. If B hoards the proceeds, the second sale need not occur. Equal aggregate stocks and quotes therefore permit different realized paths. This establishes a possibility of interrupted circulation, not a theorem that interruption or crisis is necessary.

9 Bridge to the existing mathematical-finance research

9.1 What this installment provides

The preceding models provide distinct objects for physical inventory, socially normal labor requirements, monetary expression, observed prices, and realized payments. Later financial contracts must specify which cash flows they claim, who owes them, and when they are payable. An inventory labor coefficient alone does not determine a traded claim's price or an equivalent martingale measure.

The existing draft's interpretation of Girsanov as economic redistribution requires particular care. A change of probability measure reweights events for a mathematical representation. It does not itself transfer goods, money, or surplus-value between owners. Redistribution needs cash-flow and ownership equations, such as (10), extended to production and claims.

9.2 Finite-horizon measure changes: an explicit check

The existing draft also links asymptotic divergence of a Girsanov kernel to failure of finite-horizon equivalence. The implication needs an independent proof and is false in general. Let W be Brownian motion and take the deterministic kernel $\theta_t = t$. For each finite T,

$$\mathbb{E} \exp\left(\tfrac{1}{2}\int_0^T \theta_t^2 \, \mathrm{d}t\right) = \exp(T^3/6) < \infty.$$

Thus Novikov's sufficient criterion holds on every finite horizon, and

$$Z_T = \exp\left(-\int_0^T t \, \mathrm{d}W_t - \tfrac{1}{2}\int_0^T t^2 \, \mathrm{d}t\right)$$

is strictly positive with $\mathbb{E}Z_T = 1$. It defines an equivalent measure on $\mathcal{F}_T$ even though $\theta_t \to \infty$. Infinite-horizon equivalence is a different question. Moreover, failure of a *sufficient* criterion is not proof that no equivalent measure exists. Stochastic integration and change of measure are treated in [4].

Falling profits, default, and liquidity restrictions may be important economic mechanisms, but their connection to no-arbitrage must be established for a specified market, admissible strategies, gain processes, and time horizon. A defaultable payoff is not by itself a contradiction of arbitrage-free pricing. The counterexample here repairs the logical bridge for future work; it is not a review of all claims in the earlier manuscript.

10 Open closures and the next installments

The present construction is a baseline, with several open problems. First, one must choose an interpretation of socially necessary labor: production norms, market validation, and their interaction need not produce the same v. A validation factor multiplying unsold output would be an additional theory; neither "unsold implies zero value" nor "all private hours are socially valid" has been assumed here.

Second, normal techniques and the weights α_k should eventually be linked to data or institutional mechanisms. Equation (1) specifies a reduction once weights are given; it does not determine them. Third, a money-selection model must explain acceptance rather than obtain money solely by selecting a graph root. Fourth, price wedges and sale intensities need mechanisms of demand, access, and bargaining. Those mechanisms determine whether interruptions are transient, recurrent, or systemic.

To finish the more detailed treatment of Chapter 3, add explicit dated debts $D^{ac}(T)$, separate commodity delivery from payment, and specify legally admissible netting and settlement. Hoards can then serve as payment reserves. For world money, introduce settlement domains, exchange rates, and international transfers. Converting the accepted asset to fiat or deposit money requires a balance-sheet issuer and redemption or settlement rules. These are extensions, not results already obtained by the closed cash-sale model.

The broader program follows the order of the text:

1. Volume I, Part II: distinguish C–M–C from M–C–M'; formalize labor-power as a special commodity and state the conditions under which exchange alone cannot explain an aggregate increment.

2. Volume I, Parts III–V: model the labor process, constant and variable capital, the working day, and absolute and relative surplus-value.

3. Volume I, Parts VI–VIII: wages, reproduction, accumulation, and the historical conditions of capitalist ownership. Historical arguments need evidence alongside formal models.

4. Volume II: turnover, inventories, circuit interruption, and reproduction across departments; distinguish stock and flow dimensions.

5. Volume III: profit formation, competition, commercial and interest-bearing capital, rent, and financial claims. Reassess the existing finance results against the foundations established earlier.

The next focused task is the social determination of ℓ in Chapter 1, §§1–2. Comparing a production-normalization closure with an explicit market-validation closure will expose where their predictions differ before a more elaborate stochastic system is chosen.

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